Out-of-Balance Force Calculation

Problem

M1: Two identical holes are drilled in a uniform circular disc at a radius of 400 mm from the axis. The total mass of the material removed is 187.5 g. Calculate the resultant out-of-balance force if the holes are spaced 90° apart and the disc rotates at 1000 rpm. Determine the position and magnitude of a mass that should be placed at a radius of 250 mm to balance the disc.

Solution

Given Data

- Radius of the holes: $r = 400\,\text{mm} = 0.4\,\text{m}$
- Total mass removed: $m_{\text{total}} = 187.5\,\text{g} = 0.1875\,\text{kg}$
- Mass removed per hole: $m = \frac{m_{\text{total}}}{2} = 0.09375\,\text{kg}$
- Angular velocity: $\omega = 1000\,\text{rpm} = \frac{1000\times 2\pi}{60}\,\text{rad/s} = 104.72\,\text{rad/s}$
- Radius for balancing mass: $r_b = 250\,\text{mm} = 0.25\,\text{m}$

Resultant Out-of-Balance Force

The centrifugal force for each hole is given by:

$$F = m \cdot r \cdot \omega^2$$

For two holes spaced 90° apart, the resultant force is:

$$F_{\text{resultant}} = \sqrt{F_1^2 + F_2^2} = \sqrt{2 \cdot F^2}$$

Substituting $F = m \cdot r \cdot \omega^2$:

$$F_{\text{resultant}} = \sqrt{2 \cdot (m \cdot r \cdot \omega^2)^2}$$

Simplify:

$$F_{\text{resultant}} = \sqrt{2} \cdot m \cdot r \cdot \omega^2$$

Substitute the values (**2 marks**):

$$F_{\text{resultant}} = \sqrt{2} \cdot 0.09375 \cdot 0.4 \cdot (104.72)^2$$

$$F_{\text{resultant}} \approx 580.98\,\text{N}$$

Balancing Mass

To balance the disc, the balancing mass m_b at radius r_b must produce an equal and opposite force:

$$F_b = m_b \cdot r_b \cdot \omega^2$$

Equating F_b to $F_{\text{resultant}}$:

$$m_b \cdot r_b \cdot \omega^2 = F_{\text{resultant}}$$

Solve for m_b:

$$m_b = \frac{F_{\text{resultant}}}{r_b \cdot \omega^2}$$

Substitute the values (**2 marks**):

$$m_b = \frac{580.98}{0.25 \cdot (104.72)^2} = \frac{580.98}{2737.52} \approx 0.2119\,\text{kg}$$

Angle of Opposite Force

The balancing mass must be placed at an angle opposite to the resultant force. Since the two holes are spaced 90° apart, the resultant force acts at an angle of 45° from either hole. To counteract this force, the balancing mass should be placed at an angle of:

$$\theta_b = 45^\circ + 180^\circ = 225^\circ$$

This ensures (i.e. 45° to a drilled hole) (**1 marks**): the balancing force is directly opposite to the resultant force.

Final Answer

- Resultant out-of-balance force: 580.98 N
- Balancing mass: 0.2119 kg at a radius of 250 mm
- Angle for balancing mass: 225° or 45° to a drilled hole.